MEHUL MITTAL

AI Engineer | Generative AI, Deep Learning & Production ML Systems

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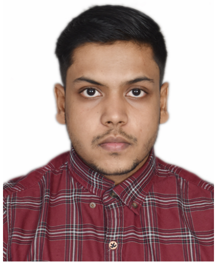

Date of Birth: 27.12.1999 | **Nationality:** Indian | **Location:** Nürnberg, Germany
Languages: English (Fluent) · German (A2, Elementary) · Hindi (Native)

PROFESSIONAL SUMMARY

AI Engineer with 4+ years of experience building production ML systems, LLM-powered applications, and agentic workflows across industry and research. Engineered an LLM-based automation pipeline at Robert Bosch with structured output parsing and prompt chaining (60% reduction in manual review) and led development of a production ML platform at Concentrix serving 200+ users (€2M ROI). Published researcher in healthcare AI (IEEE-EMBS) with hands-on expertise in PyTorch, LangChain, RAG architectures, AWS infrastructure, and end-to-end MLOps. M.Sc. in Artificial Intelligence from FAU Erlangen-Nürnberg.

WORK EXPERIENCE

Mar 2025 – Dec 2025

ML Infrastructure Engineer (Working Student)

Robert Bosch GmbH | Abstatt, Germany

Focus: GenAI, LLM Infrastructure & Cloud MLOps

- Engineered a multi-step LLM pipeline with prompt chaining, structured output parsing, and tool integration (LangChain, Hugging Face) to automate technical diagram standardization, eliminating 60% of manual review cycles.
- Built and deployed cloud ML components on AWS (SageMaker, EC2, Lambda), contributing to distributed training and inference infrastructure serving CV and NLP models for global engineering teams.
- Implemented MLOps automation using GitHub Actions and Docker for continuous training, model versioning, and A/B evaluation pipelines, increasing deployment frequency by 3×.
- Optimized inference deployments through quantization and mixed-precision serving (ONNX, TensorRT), reducing infrastructure costs by 25% while maintaining latency SLAs.

Mar 2024 – Apr 2025

AI Research Engineer (Working Student)

Machine Learning & Data Analytics Lab, FAU | Erlangen, Germany

Focus: Healthcare AI, Time-Series Analysis & Signal Processing

- Designed and trained deep learning models (CNNs, RNNs, Transformers) in PyTorch for real-time freeze of gait detection in Parkinson's patients, processing multi-modal wearable IMU sensor data.
- Built real-time inference pipelines with signal processing (Wavelet transforms, FFT) and automated feature extraction, achieving sub-second prediction latency on edge-constrained hardware.
- Developed transfer learning and domain adaptation approaches (self-supervised pre-training, adversarial alignment) for cross-subject generalization, improving baseline F1 by 12%.
- Co-authored IEEE-EMBS 2025 conference paper on subjective motor activity perception and gait patterns in Parkinson's disease using wearable sensor analytics

Oct 2021 – Feb 2023

Lead AI/ML Engineer

Concentrix | Gurgaon, India

Focus: Production ML at Scale, Leadership & Business Intelligence

- Led development of a production ML platform (Scikit-learn, XGBoost, PyTorch) serving 200+ users with real-time predictions, ensemble methods, and automated feature engineering.
- Built time-series forecasting and anomaly detection models using gradient boosting and neural networks, delivering €2M ROI through optimization algorithms.
- Implemented model monitoring with drift detection (Evidently AI), explainability dashboards (SHAP, LIME), and automated retraining pipelines.
- Mentored 5 junior ML engineers on deep learning fundamentals, neural network architectures, and production ML best practices.

Sep 2021 – Nov 2021

ML Engineering Intern

Mirrag AI | Mumbai, India

Focus: Computer Vision & MLOps Foundation

- Developed object detection and image classification models (YOLO, ResNet, EfficientNet) using PyTorch and transfer learning, achieving 85%+ accuracy on production datasets.
- Established MLOps foundation with experiment tracking (MLflow, W&B), data versioning (DVC), and containerization (Docker), reducing experimentation cycle time by 40%.

EDUCATION

Apr 2023 – Jan 2026

M.Sc. in Artificial Intelligence

Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU) | Erlangen, Germany

Specialisation: Deep Learning, Computer Vision, NLP, Reinforcement Learning, Probabilistic ML, MLOps

Thesis: Subject-Invariant EEG Representation Learning (Cross-Subject Generalisation with Deep Learning)

Awarded FAU Graduation Scholarship (Dec 2025) for academic excellence in AI research

Aug 2018 – Jun 2022

B.Tech in Information Technology

Guru Gobind Singh Indraprastha University | New Delhi, India

Capstone: Team Lead for Distributed ML System Implementation

TECHNICAL SKILLS

Category	Technologies & Tools
Programming	Python, SQL, R, C++, JavaScript, Bash, REST API Design (FastAPI)
GenAI & LLMs	GPT-4, Claude API, Llama, LangChain, LangGraph, LlamaIndex, LoRA/QLoRA, Prompt Engineering, Structured Outputs, Tool Calling
Agentic & RAG	Agentic Workflows, RAG Pipelines, Semantic Search, Pinecone, ChromaDB, FAISS, Weaviate, MCP
Deep Learning	PyTorch, TensorFlow, Keras, PyTorch Lightning, Hugging Face Transformers, Vision Transformers (ViT)
Machine Learning	Scikit-learn, XGBoost, LightGBM, CatBoost, Optuna, Ray Tune, SHAP, LIME
Computer Vision	OpenCV, YOLO, Faster R-CNN, ResNet, EfficientNet, Semantic Segmentation
NLP	spaCy, NLTK, BERT, Sentence Transformers, Word2Vec, GloVe, NER, Text Classification
MLOps & Eval	MLflow, Weights & Biases, DVC, Docker, Kubernetes, GitHub Actions, TorchServe, LLM Evaluation Pipelines
Cloud & Infra	AWS (SageMaker, Lambda, EC2), Azure ML, Databricks, Google Cloud AI Platform
Optimisation	Quantization, ONNX, TensorRT, Knowledge Distillation, Mixed Precision Training

KEY PROJECTS

Oct 2024 – Apr 2025

Healthcare Analytics with Large Language Models

Academic Capstone | FAU Erlangen-Nürnberg

- Built GenAI pipeline integrating GPT-4 and Claude with RAG architecture (Pinecone, ChromaDB) and semantic vector search for medical knowledge retrieval.
- Implemented automated monitoring for LLM drift detection, hallucination metrics, and MLOps governance with MLflow 3.0 meeting healthcare compliance standards.

May 2025 – Nov 2025

Neural Transfer Learning for Cross-Subject Generalisation

Master's Thesis | FAU Erlangen-Nürnberg

- Built end-to-end deep learning pipeline processing 1M+ EEG samples with automated augmentation using PyTorch Lightning; benchmarked EEGNet, EEGConformer, ATCNet, and Vision Transformers with domain adaptation (DANN, contrastive learning).
- Deployed reproducible research infrastructure with MLflow experiment tracking, model registry, and DVC dataset versioning.

PUBLICATIONS

Slim, S., Küderle, A., Moradi, H., **Mittal, M.**, Salin, E., Winkler, J., & Eskofier, B. (2025). *Investigating Subjective Motor Activity Perception and Gait in Parkinson's Disease.* IEEE-EMBS International Conference on Biomedical and Health Informatics.